



Model-agnostic Local Feature Importance Methods

Feature Importance

Lecture: Explainable Artificial Intelligence, ST 2026

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Outline

Evaluating Local Feature Importance Methods

Black-box Local Feature Importance Methods

- Local Interpretable Model-agnostic Explanations (LIME)

- RISE

- Shapley-values (SHAP)

From local to global explanations

Summary

Exam Announcement

The exam will take place as **written exam**.

- ▶ 90 min
- ▶ Semi-open book: 2 DIN A4 pages personal cheat sheet allowed
- ▶ when & where tba

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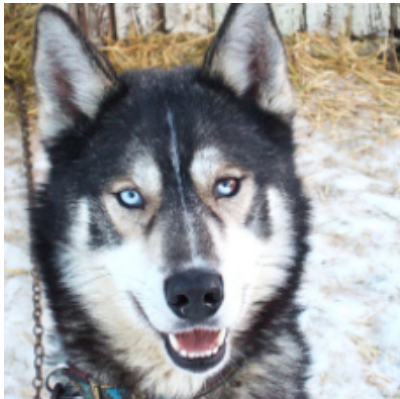
Local Feature Importance

Which input features were relevant this output?

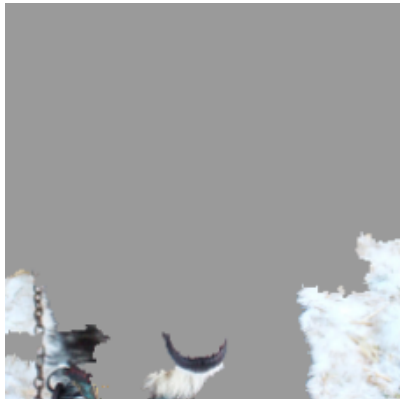
Key differences

- ▶ What is a *feature*?
- ▶ When is a feature (more) *relevant* (than another)?

Why is it hard to evaluate attribution?



***Husky* image**
misclassified as *Wolf*



features most influential
for the decision

What makes good feature importance?

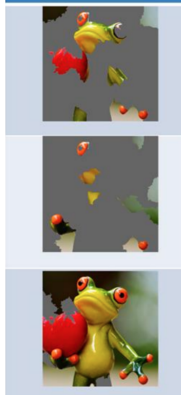
Capture features that, if removed, *change* the output / performance.

- ▶ *Recall* on known-to-be-relevant features [10]: *On transparent models* recover truly used features
- ▶ *Monotonicity* [9]: *Upon varying a (single) feature*: its relevance correlates to expected change in loss.
- ▶ *non-sensitivity* [9]: 0-relevance reflects functional independence
- ▶ *Sensitivity-n* [1] / *infidelity* [11]: *Upon removal of several features*: (accumulated) relevance reflects output variance / change
- ▶ *Explanation selectivity* [8]: *Upon removal from most to least relevant*: model confidence drops sharply

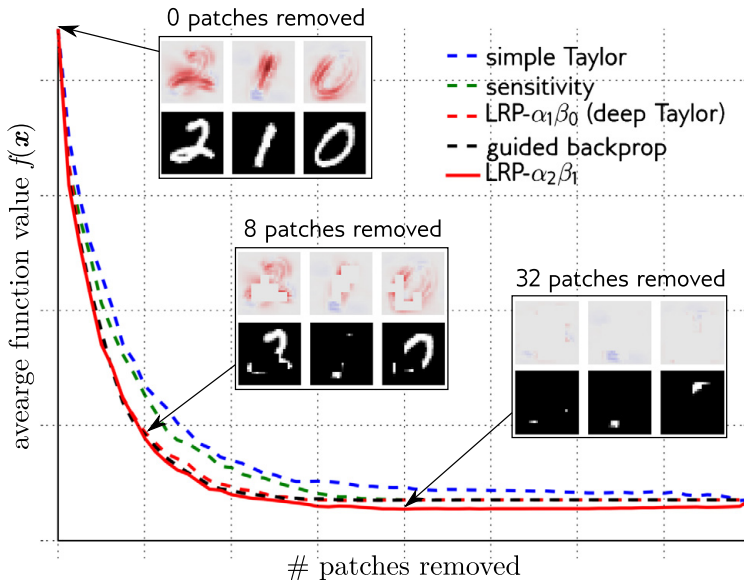
Other:

- ▶ *Explanation continuity* [8, Sec. 7.2]: *Robustness* to (too) small perturbations

Perturbed Instances



Explanation Selectivity



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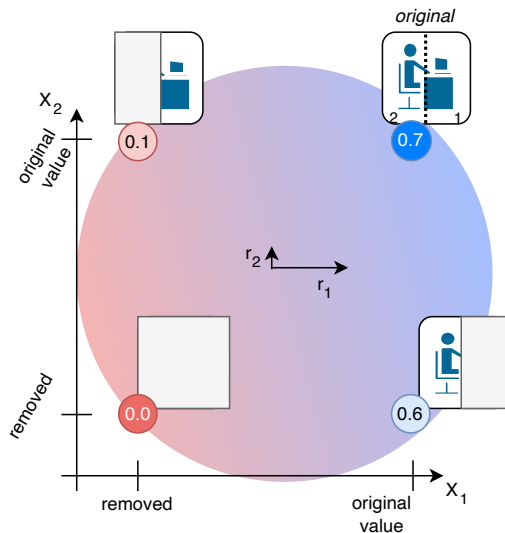
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General Local Surrogates

Idea: Generate an explainable surrogate model of the black-box's behavior in the *vicinity* around the explanandum (sample). Derive an explanation for the explanandum from the surrogate.

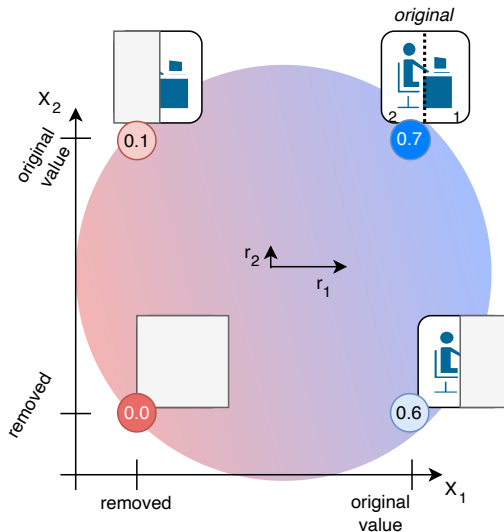
Idea of Local Interpretable Model-agnostic Explanations (LIME) [10] $\Rightarrow$



Generic Local Surrogate Algorithm

Given instance x and black-box f ,

1. **Local behavior:** Perturb x to generate new dataset $\tilde{\mathbf{d}}(x) = ((\tilde{x}^{(i)}, f(\tilde{x}^{(i)})))_{i \leq n}$
2. **Weight** new samples according to their proximity π_x to x .
3. **Train surrogate** g_x to fit weighted data (e.g., linear regressor).
4. **Explain** $f(x)$ via g_x .



LIME Algorithm

1. **Local features:** e.g., tabular features, image segmentation, ...
2. **Perturbation:** Consider features as **binary** (original vs. removed), assume $\text{enc}_x: \{1, 0\}^K \rightarrow X$ s.t. $\text{enc}(z) = x$ for $z = (1, \dots, 1)$; $\tilde{\mathbf{d}}(z) \subset \{0, 1\}^K$ randomly sampled
3. **Surrogate:**
minimize the weighted faithfulness error and complexity

$$g_x^* = \underset{g \in G}{\operatorname{argmin}} \sum_{\tilde{z} \in \tilde{\mathbf{d}}(x)} \overbrace{\pi_x(\tilde{x})}^{\text{stay in vicinity}} \underbrace{(f(\text{enc}(\tilde{z})) - g(\tilde{z}))^2}_{\text{faithfulness error}} + \underbrace{\Omega(g)}_{\text{complexity}}$$

where, e.g., $\pi_x(\tilde{z}) = 1 - \frac{\#\text{zeros}(\tilde{z})}{K}$

Perturbed Instances



How to remove a feature?

Common goal: eliminate information about the feature value.

Some typically used examples are:

Tabular data: *feature* = column; *removal* = randomize

Natural language text: *feature* = word; *removal* = remove word from text

Images: *feature* = superpixel;



Original Image



Interpretable
Components

removal = set superpixel to

- ▶ uniform average (gray)
- ▶ blur
- ▶ noise (bad idea)

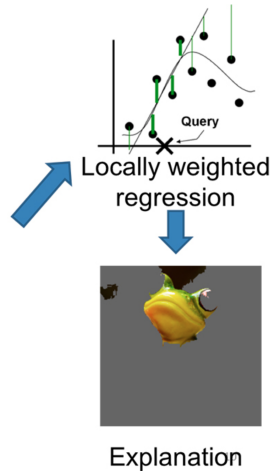
LIME: Image Example



Original Image
 $P(\text{tree frog}) = 0.54$



Perturbed Instances	$P(\text{tree frog})$
	0.85
	0.00001
	0.52



Pros and Cons of LIME

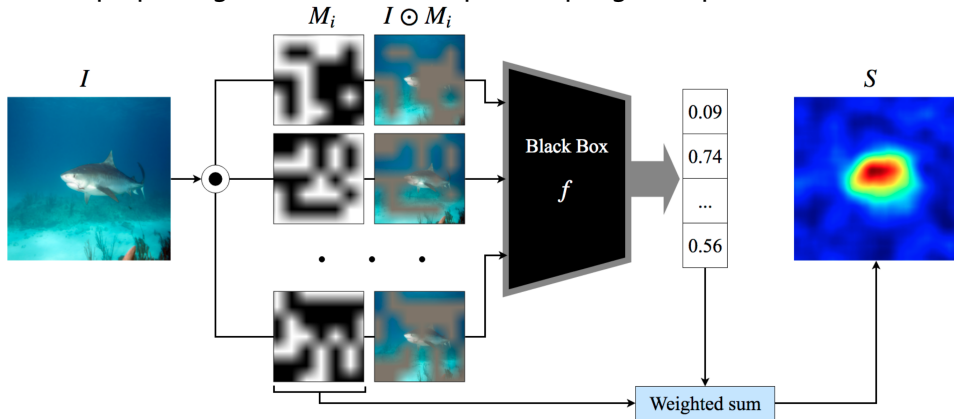
- + Very accessible & popular (good implementation)
- Highly depends on
 - ▶ selected features,
 - ▶ encoding scheme enc,
 - ▶ distance metric π_x ;may result in unrealistic inputs.
- Only binary feature removal.
- **Costly**: n model evaluations needed!

Perturbed Instances



Simplifying: RISE

Discarding similarity weights and using random masks instead of combinations of superpixels gives Randomized Input Sampling for Explanation (RISE).



Inspiration from cooperative game theory

How should the **payoff** $\nu(S)$ (impact to output)
as generated by different **coalitions** $S \subset \mathbf{S}$ (bin. feature combinations $\in \{0, 1\}^K$)
of self-interested **agents** (features)
be **distributed** amongst the team members? (individual feature relevance)

Example

- ▶ How to distribute usage time to people who collaboratively bought a tool?
- ▶ How to distribute payment to craftsmen constructing a building?
- ▶ How should km of the taxi ride home be accounted to passengers $\in \mathbf{S} = \{A, B, C\}$?
 $\nu(\{A\}) = 6, \nu(\{B\}) = 12, \nu(\{C\}) = 42,$
 $\nu(\{A, B\}) = 12, \nu(\{A, C\}) = 42, \nu(\{B, C\}) = 42, \nu(\{A, B, C\}) = 42$

Shapley Values

Idea: every member $i \in S$ should receive a value $\Psi_i(S, \nu)$ proportional to their contribution.

Definition (Shapley Fairness)

A fair Ψ should be:

efficient & feasible: $\nu(S) \leq \sum_{i \in S} \Psi_i(S, \nu) \leq \nu(S) \quad \forall S$;

symmetric: $\nu(S \cup \{i\}) = \nu(S \cup \{j\}) \forall S \subset N \setminus \{i, j\} \implies \Psi_i(\bullet, \nu) = \Psi_j(\bullet, \nu)$

dummy player aware: $\nu(S \cup \{i\}) = \nu(S) + \nu(\{i\}) \forall S \subset N \setminus \{i\} \implies \Psi_i(\bullet, \nu) = \nu(\{i\})$

additive: For two payoffs ν_1, ν_2 : $\Psi_\bullet(S, \nu_1 + \nu_2) = \Psi_\bullet(S, \nu_1) + \Psi_\bullet(S, \nu_2)$

Theorem (Shapley Values)

$\phi_i = \Psi_i(S, \nu) := \frac{1}{|S|} \sum_{S \subset S \setminus \{i\}} \underbrace{\left(\binom{|S|-1}{|S|} \right)^{-1}}_{\text{\# coalitions of that size}} \underbrace{(\nu(S \cup \{i\}) - \nu(S))}_{\text{value added by } i \text{ to } S}$ uniquely fulfills above.

From Shapley Values to SHAP

Back to explanations, recall that we want to answer

How should the **predicted value** $f(\text{enc}(\tilde{z}))$ (value $\nu(\tilde{z})$)
as generated by different **feature combinations** $\tilde{z} \in \{0, 1\}^K$ (coalitions)
of **feature values** of x (agents)
be accounted to **feature relevancies**? (distribution amongst team members)

$\Rightarrow$ Shapley value ϕ_i is relevancy of feature i !

However, calculating outputs $f(\text{enc}(\tilde{z}))$ for all feature combinations $\tilde{z}$ is infeasible (exponential runtime!).

From Shapley Values to SHAP

SHapley Additive exPlanations (SHAP)-based estimation of Shapley values [6]:

- ▶ KernelSHAP (**costly**): Use LIME with weight $\pi_x(\tilde{z}) = \frac{(K-1)}{|\tilde{z}| \binom{K}{|\tilde{z}|} (M-|\tilde{z}|)}$,
then the solutions ϕ_i to $g(\tilde{z}) = \mathbb{E}_{\tilde{z}'} [f(\text{enc}(\tilde{z}'))] + \sum_{i=1}^K \phi_i \tilde{z}_i$ are the Shapley values.
- ▶ Permutation: consider only *sequences* with one permutation at a time,
e.g., add A, B, then C ($\{A\}, \{A, B\}, \{A, B, C\}$); *and backwards* ($\{C\}, \{B, C\}, \{A, B, C\}$).
- ▶ CorrSHAP [4]: only consider coalitions of correlated features
- ▶ Architecture specific versions, *e.g., TreeSHAP*

Pros and Cons

- + Very principled method with several nice theoretical properties.
- Still heavily depends on the choice of features
- Relies on (costly) approximation
- KernelSHAP: Also suffers from possibly unrealistic samples obtained via perturbation.

Outlook:

- ▶ Also works for plotting dependence plots.
- ▶ Shapley values come with a mathematical definition of Interaction Values.
- ▶ Efficiency is active line of research, amongst others in our institute! ^[4]

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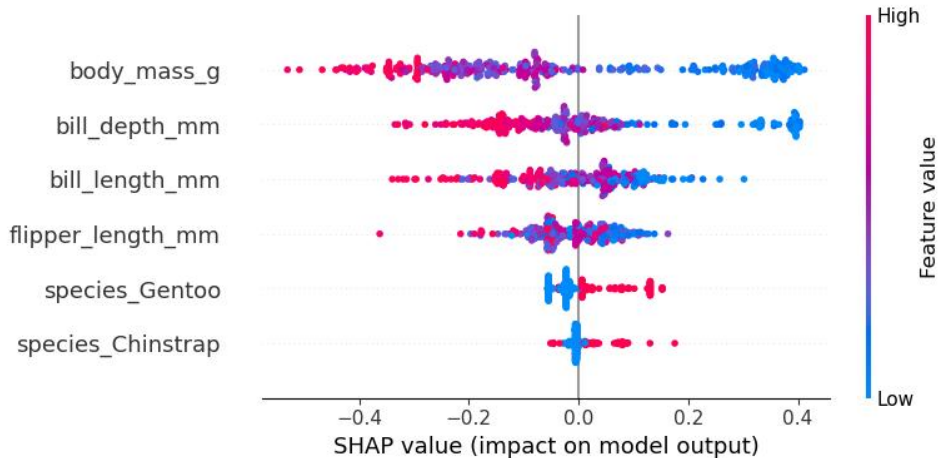
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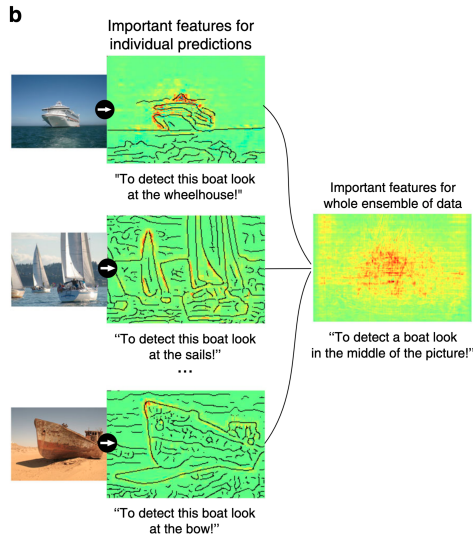
Local-to-global via Summary Plots



Local-to-global via Clustering: SpRAy

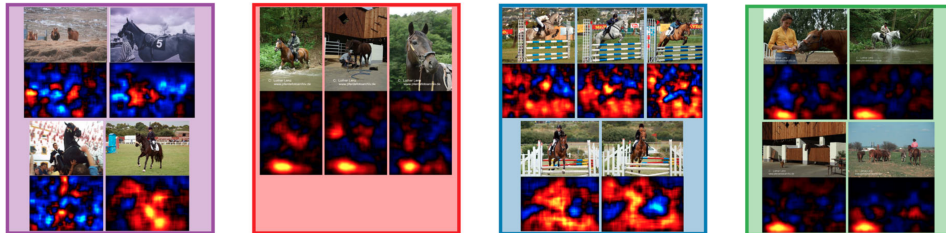
Idea (for *the example of SpRAy* [5]):

- ▶ Collect **local feature importance** vectors,
e.g., from layerwise relevance propagation.
- ▶ Apply **clustering**: distinct clusters correspond to distinct strategies,
e.g., use spectral clustering.
- ▶ **Analyze** cluster structure: far-away clusters indicate very different strategies,
e.g., indicated by large gap in eigenvalues.



[5, Fig. 1]

Local-to-global via Clustering: SpRAy

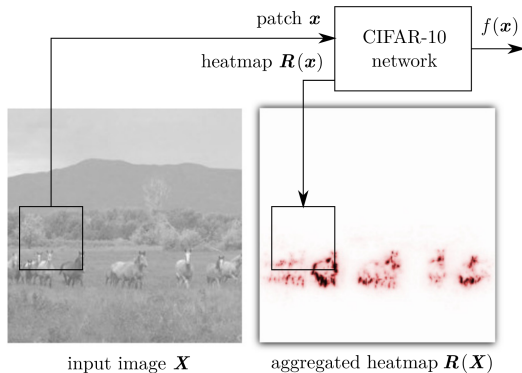


⇒ yields distribution information for free!

Patch-to-global Example: Windowing

What to do in case only heatmaps of (overlapping) patches are available?

⇒ sum relevance per pixel.



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- ▶ Common concept: **perturbing samples** via true removal, uniform/blur, random
- ▶ **LIME** (or general perturbation-based feature importance methods): Switch features on and off and weight newly obtained samples, then fit explainable model.
- ▶ **SHAP**: aims to approximate Shapley values of features = fair distribution of relevance according to individual feature's contribution to the output value in any of its feature combinations
- ▶ From **local to global**: E.g., use summary plots or clustering
- ▶ Common issues: Lots of model evaluations needed, highly dependend on choice of features

Further Reading

- ▶ Metrics: Zhou et al., “Evaluating the Quality of Machine Learning Explanations” [12], Chap. 6.3
- ▶ LIME: Christoph Molnar, *Interpretable Machine Learning* [2], Chap. 14
- ▶ SHAP: Christoph Molnar, *Interpretable Machine Learning* [2], Chap. 17–18;
- ▶ SHAP deep dive: Molnar, *Interpreting Machine Learning Models With SHAP* [7]
- ▶ Other black-box local feature importance techniques: Guidotti et al., “A Survey of Methods for Explaining Black Box Models” [3], Sec. 7.2

References I

- [1] Marco Ancona et al. "Towards Better Understanding of Gradient-Based Attribution Methods for Deep Neural Networks". In: *International Conference on Learning Representations*. Feb. 2018. (Visited on 05/05/2026).
- [2] Christoph Molnar. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. 3rd ed. 2025.
- [3] Riccardo Guidotti et al. "A Survey of Methods for Explaining Black Box Models". In: *ACM Comput. Surv.* 51.5 (Aug. 2018), 93:1–93:42. ISSN: 0360-0300. DOI: 10.1145/3236009 [↗](#). arXiv: 1802.01933 [↗](#). (Visited on 08/18/2019).
- [4] Vahidin Hasić, Amar Halilović, and Senka Krivić. "Superpixel Correlation for Explainable Image Classification". In: *Explainable Artificial Intelligence*. Ed. by Riccardo Guidotti, Ute Schmid, and Luca Longo. Cham: Springer Nature Switzerland, 2026, pp. 27–44. ISBN: 978-3-032-08330-2. DOI: 10.1007/978-3-032-08330-2_2 [↗](#).
- [5] Sebastian Lapuschkin et al. "Unmasking Clever Hans Predictors and Assessing What Machines Really Learn". In: *Nature Communications* 10.1 (Mar. 2019), p. 1096. ISSN: 2041-1723. DOI: 10.1038/s41467-019-08987-4 [↗](#). (Visited on 03/24/2021).
- [6] Scott M Lundberg and Su-In Lee. "A Unified Approach to Interpreting Model Predictions". In: *Advances in Neural Information Processing Systems* 30. Ed. by I. Guyon et al. Long Beach, CA, USA: Curran Associates, Inc., 2017, pp. 4765–4774. arXiv: 1705.07874 [↗](#). (Visited on 04/30/2019).
- [7] Christoph Molnar. *Interpreting Machine Learning Models With SHAP*. Dec. 2025. (Visited on 05/06/2026).
- [8] Grégoire Montavon, Wojciech Samek, and Klaus-Robert Müller. "Methods for Interpreting and Understanding Deep Neural Networks". In: *Digital Signal Processing* 73 (Feb. 2018), pp. 1–15. ISSN: 1051-2004. DOI: 10.1016/j.dsp.2017.10.011 [↗](#). (Visited on 12/09/2019).
- [9] An-phi Nguyen and María Rodríguez Martínez. *On Quantitative Aspects of Model Interpretability*. July 2020. DOI: 10.48550/arXiv.2007.07584 [↗](#). arXiv: 2007.07584 [cs] [↗](#). (Visited on 05/05/2026).
- [10] Marco Túlio Ribeiro, Sameer Singh, and Carlos Guestrin. "Why Should I Trust You?: Explaining the Predictions of Any Classifier". In: *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*. KDD '16. San Francisco, California, USA: ACM, 2016, pp. 1135–1144. ISBN: 978-1-4503-4232-2. DOI: 10.1145/2939672.2939778 [↗](#).

References II

- [11] Chih-Kuan Yeh et al. "On the (In)Fidelity and Sensitivity of Explanations". In: *Advances in Neural Information Processing Systems*. Vol. 32. Curran Associates, Inc., 2019. (Visited on 05/05/2026).
- [12] Jianlong Zhou et al. "Evaluating the Quality of Machine Learning Explanations: A Survey on Methods and Metrics". In: *Electronics* 10.5 (Jan. 2021), p. 593. DOI: 10.3390/electronics10050593 [↗](#).